

Digital Mice: Inducing and Validating Controlled Cognitive Deficits in Frontier Language Models for Mechanism Design

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Abstract

Early-stage mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must run experiments on people before they can learn which mechanisms are robust to bounded rationality. We ask whether frontier reasoning models can serve as the economist’s analogue of the laboratory mouse—capable reasoners instructed into specific, controlled cognitive deficits whose behavior can be studied before human subjects are recruited. Using a canonical second-price sealed-bid auction (SPSB) and a $3 \times 3 \times 3$ panel of *digital mice* defined over three cognitive axes (contingent reasoning, forward planning, belief reasoning), we establish three facts. First, the deficits have *construct validity*: repairing each axis moves truthful bidding in the theory-predicted direction, monotonically and with the predicted sign in all five pre-registered directional tests. Second, the panel—augmented by a small set of targeted error strains—attains *calibration coverage*: it reproduces all five canonical SPSB human-mistake classes (truthful, moderate underbidding, overbidding, extreme overbidding, and near-zero bidding) at rates above their detection thresholds. Third, the panel is *discriminating*: scored against candidate auction mechanisms under two base models, it exposes a cross-model portability gap in which an all-pay design’s intended “bid aggressively” behavior collapses on a stronger held-out model while a second-price design’s truthful target stays stable. These results support the use of instructed-deficit language models as a phase-I screening population for mechanism design.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity.

1 Introduction

Medicine studies humans, but it does not begin with them. Much early drug testing starts with mice, and some of those mice are deliberately engineered to lack a specific protein, receptor, or channel. The deficit is not a nuisance; it is the identification strategy. If a treatment succeeds or fails only when a particular channel is removed, the experiment isolates a mechanism that is entangled in an intact organism.

Economics has lacked this luxury. The behavior of interest in mechanism design is human cognition—contingent reasoning, forward planning, belief formation, and strategic sophistication—and these channels are bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Natural and laboratory experiments rarely deliver a population missing exactly one cognitive module, so the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models change the experimental object. The strongest current reasoning models are, in simple games, often closer to a fully rational agent than an ordinary subject: they follow long chains of reasoning and compute through cases. That capability is precisely what makes them useful here. We start from a capable reasoning substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation, not through dependence on hidden quirks of one model. This yields *digital mice*: language-model agents with specific, engineered cognitive deficits.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality states that (i) produce the behavioral signatures predicted by cognitive theory and (ii) span the empirically documented space of human mistakes, in a canonical second-price sealed-bid auction?

Contribution. We answer both halves affirmatively for the SPSB environment. Over a $3 \times 3 \times 3$ panel of 27 strains (810 bidder-decisions), repairing each cognitive axis raises the truthful-bid rate by +0.23 (contingent reasoning), +0.14 (forward planning), and +0.05 (belief reasoning); all five pre-registered directional hypotheses pass their sign test. The panel, augmented by two targeted error strains, covers all five canonical SPSB human-mistake classes at rates above threshold. Third, using the panel as an evaluator of four

candidate auction mechanisms under two base models, we find robustness does not fully port: the all-pay designs’ intended “bid aggressively” rate falls from roughly 0.65 to 0.25 on a stronger held-out model, while a second-price design’s truthful target is unchanged. We are explicit about a boundary of the construction: the clean procedural basis naturally produces truthful and underbidding behavior but not overbidding, which we recover with a small number of hand-specified heuristic strains. We treat the digital-mouse panel as phase-I screening evidence, not as a prediction of human population shares.

2 The Digital-Mouse Panel

2.1 Construction: a rational base plus a deficit instruction

A digital mouse is a single prompt assembled from two layers. The first layer, the *rational base*, is identical across all strains and instructs the model to play the auction as a long-run profit maximizer:

*Your TOP PRIORITY is to place bids which maximize your profit in the long run.
Learn from the history of previous rounds in order to maximize your total profit.
Don’t forget the values are redrawn independently each round.*

The second layer, *defect induction*, appends a “cognitive style” block containing exactly one paragraph per cognitive axis. Each paragraph constrains the model’s deliberation *procedure*—not its objective—at one of three levels. The intervention is therefore a reasoning-procedure control rather than a payoff or preference change. Concretely, the contingent-reasoning axis ranges from a level-0 paragraph that forbids reasoning about opponents to a level-2 paragraph that mandates a worst-case/dominance check:

Contingent reasoning, level 0 (deficit). *Do NOT enumerate possible bids by other bidders or compute best responses to specific opponent moves. Choose your bid without conditioning on what others might do.*

Contingent reasoning, level 2 (repaired). *For each possible bid you could make, consider the worst-case outcome given the range of possible opponent bids. Find the bid where even the worst-case outcome is acceptable. If you deviated from this bid, could the worst case become worse?*

The forward-planning and belief-reasoning axes are constructed the same way (from “do not trace consequences” to a multi-step decision tree, and from “do not model others” to explicit

Table 1: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; the lowest level is the engineered deficit.

Axis	Level 0 (deficit)	Level 1	Level 2 (repaired)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

second-order beliefs). A strain is thus the rational base plus a triple $(c, f, b) \in \{0, 1, 2\}^3$ that selects one paragraph for each axis. The all-off triple $(0, 0, 0)$ is the most deficient mouse; the all-repaired triple $(2, 2, 2)$ is closest to the rational base.

2.2 The $3 \times 3 \times 3$ grid

Varying three axes at three levels each gives the 27-strain panel of Table 1. A useful intuition: the rational base supplies the *goal* (maximize profit), and the cognitive-style block rations the *reasoning steps* the mouse is allowed to use to pursue it. A mouse told not to enumerate opponents, not to trace payoffs, and not to model others must fall back on a crude own-value heuristic; a mouse granted all three faculties can in principle reconstruct the dominant strategy.

Hardening the base (strict variant). The base layer above makes profit maximization the top priority, which leaves room for a very capable model to “see through” a deficit and revert to the optimal strategy. For the cross-model analysis of Section 6 we therefore also build a *strict* variant whose base layer inverts the priority—“follow the cognitive procedure exactly . . . if the cognitive procedure conflicts with the profit-maximizing strategy, the cognitive procedure wins”—and explicitly forbids invoking dominant-strategy or Vickrey shortcuts. The strict variant is used only where noted; all construct-validation and calibration results use the standard panel.

3 Experimental Design and Methodology

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. Under standard preferences truthful bidding is a weakly dominant strategy, which makes truthfulness a clean construct-validity target: a more capable mouse should bid closer to its value.

Strains and runs. We instantiate all 27 basis strains with the same base model (gpt-5.4-mini, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total). Because the clean procedural basis does not naturally generate *overbidding*, we additionally author a small set of targeted *error mice*—heuristic strains designed to express one documented mistake each (e.g. a “second-price overgeneralizer” that treats bidding above value as safe because the winner pays the runner-up bid, and a “payment panic” strain that bids near zero). Seven error strains are specified; two had completed runs at the time of this analysis and feed the calibration scorecard below.

Behavioral classes. For each bidder-decision we record the bid, the value, and the deviation $b - v$. We define a truthful bid as $|b - v| \leq 0.05$, moderate underbidding as $b < v - 0.05$ (excluding near-zero bids), overbidding as $b > v + 0.05$, extreme overbidding as $b - v \geq 5$, and near-zero bidding as $b \leq 0.1$ with $v > 5$. These thresholds operationalize the five canonical SPSB mistake classes used in the calibration step.

3.1 Outcome variables and the robustness objective

We now define the panel-based objective used to score mechanisms in Section 6. Fix a mechanism and a persona panel $\mathcal{P} = \{p_1, \dots, p_M\}$, and run R independent auctions per persona. Index the resulting completed auctions by $a \in \mathcal{A}$, with $\mathcal{A}_p$ the subset run under persona p . Each auction has $N = 3$ bidders with private values $v_{a,i}$ and elicited bids $b_{a,i}$; the winner is $i^*(a) = \arg \max_i b_{a,i}$ and the mechanism’s pricing rule sets the payment $\rho(a)$ (the second-highest bid under SPSB, the own bid under all-pay, and so on). Let A_{att} be the number of *attempted* auctions and F the number that fail to parse or violate the response format, so the failure rate is $\phi = F/A_{\text{att}}$.

Per-auction quantities. Write $v_a^{\max} = \max_i v_{a,i}$. We use allocative efficiency, a bid-shading dispersion, and realized winner surplus:

$$\text{eff}(a) = \frac{v_{a,i^*(a)}}{v_a^{\max}}, \quad s(a) = \text{std}_{i: v_{a,i} > 0} \left(\frac{b_{a,i}}{v_{a,i}} \right), \quad \pi(a) = v_{a,i^*(a)} - \rho(a). \quad (1)$$

Panel aggregates. With $A = |\mathcal{A}|$ and a value scale $\bar{v} = \max_a v_a^{\max}$, define normalized revenue, mean efficiency, a simplicity proxy, and normalized surplus variance:

$$\hat{\rho} = \frac{1}{A\bar{v}} \sum_a \rho(a), \quad \overline{\text{eff}} = \frac{1}{A} \sum_a \text{eff}(a), \quad \sigma = 1 - \frac{1}{A} \sum_a s(a), \quad \widehat{\text{Var}} = \frac{\text{Var}_a(\pi(a))}{\bar{v}^2}. \quad (2)$$

The simplicity proxy σ is high when bidders converge on a common bid/value ratio (an “obvious” mechanism) and low when their shading scatters.

Intended-action rate. Each mechanism declares the action a correctly reasoning bidder should take, $\tau \in \{\text{TRUTHFUL}, \text{SHADE}, \text{AGGRESSIVE}, \dots\}$, with tolerance $\epsilon = 0.1$. For a scoreable decision (v, b) with $v > 0$ define the match indicator

$$m_\tau(v, b) = \begin{cases} \mathbf{1}[|b - v| \leq \epsilon] & \tau = \text{TRUTHFUL}, \\ \mathbf{1}[\kappa < b < v - \epsilon] & \tau = \text{SHADE}, \\ \mathbf{1}[b + \epsilon \geq \alpha v] & \tau = \text{AGGRESSIVE}, \end{cases} \quad (3)$$

with floor κ and aggressiveness threshold $\alpha = 0.4$ by default. The intended-action rate μ_τ is the mean of m_τ over scoreable decisions; targets labeled UNKNOWN or MECHANISM-SPECIFIC are not scored (μ_τ omitted).

Lower-tail (per-persona) statistics. Let $\overline{\text{eff}}_p$, σ_p , and $\text{Var}_p(\pi)$ be the above computed within persona p . Robustness rewards good *worst-case* behavior across the panel, not just the average:

$$E_{\min} = \min_p \overline{\text{eff}}_p, \quad E_{20} = Q_{0.2}(\{\overline{\text{eff}}_p\}), \quad \Sigma_{\min} = \min_p \sigma_p, \quad \hat{V}_{\max} = \frac{\max_p \text{Var}_p(\pi)}{\bar{v}^2}, \quad (4)$$

where $Q_{0.2}$ is the 20th percentile across personas.

Fitness. The aggregate and robust fitness are weighted sums

$$F_{\text{agg}} = w_1 \hat{\rho} + w_2 \overline{\text{eff}} + w_3 \mu_\tau + w_4 \sigma - w_5 \widehat{\text{Var}} - w_6 \phi, \quad (5)$$

$$F_{\text{tail}} = w_7 E_{\min} + w_8 E_{20} + w_9 \Sigma_{\min} - w_{10} \hat{V}_{\max}, \quad (6)$$

$$F_{\text{robust}} = F_{\text{agg}} + F_{\text{tail}}. \quad (7)$$

Following the Stage-C calibration mapping, the default weights are $(w_1, \dots, w_{10}) = (1, 1, 0.5, 0.5, 0.25, 1, 0.5, 0.5, 0.25, 0.25)$, grouping into domain performance (w_1, w_2) , intended action (w_3) , simplicity (w_4, w_9) , regret (w_5, w_{10}) , compliance (w_6) , and cognitive fragility (w_7, w_8) . Section 6 reports F_{robust} and μ_τ ; the sign convention is that fragility, regret, and failure enter negatively, so a mechanism scores well only if it is allocatively efficient, induces its intended action, and does so *uniformly* across the digital-mouse panel.

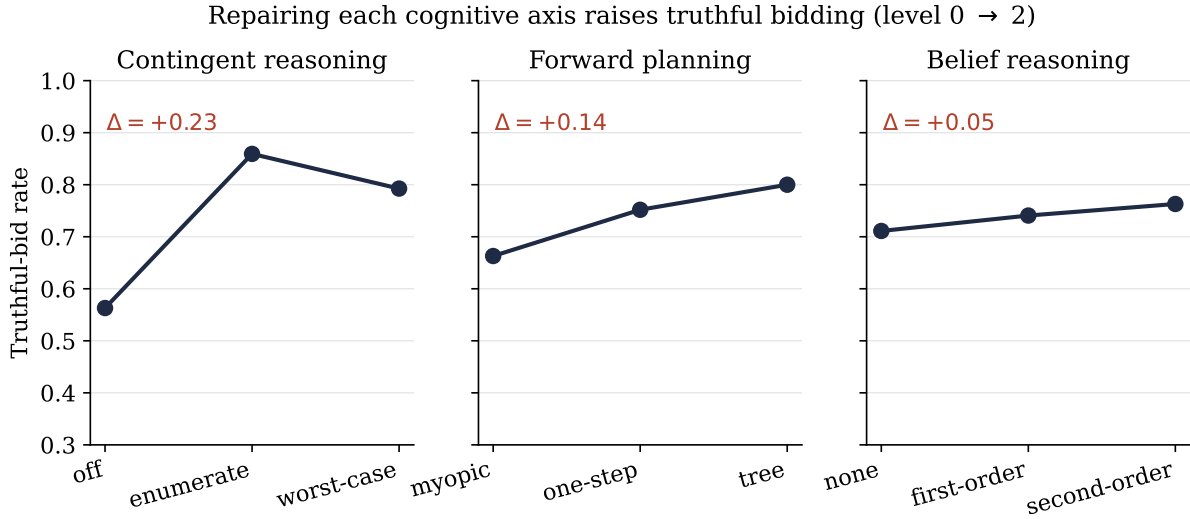


Figure 1: Construct-validity marginals. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Repairing each axis raises truthfulness, with the largest effect on the contingent-reasoning axis.

4 Result 1: Construct Validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then repairing an axis should move behavior toward the rational benchmark. Figure 1 plots the truthful-bid rate at each axis level, marginalizing over the other two axes. All three axes are monotone or near-monotone in the repairing direction, and the contingent-reasoning axis produces the largest swing.

Table 2 reports the five pre-registered directional tests. For each axis we test that repairing it raises the truthful rate (expected positive) and, where applicable, lowers the underbidding rate (expected negative). All five tests pass: the realized sign matches the prediction and exceeds the minimum-magnitude bar in every case. The directional structure is exactly what theory predicts—more contingent reasoning and more forward planning push the agent toward its dominant strategy—which is the core construct-validity claim.

Figure 2 shows that the strains are also *separable* at the level of raw behavior. The rational mouse (contingent reasoning engaged) bids on the 45° truthful line; the all-deficits-off mouse systematically underbids; and the targeted error mice occupy distinct regions—persistent overbidding above the line and near-zero bidding along the floor. The mean absolute deviation falls from 2.78 for the most deficient basis strain to essentially zero (0.003) for the most rational one.

Table 2: Construct-validity diagnostics: five pre-registered directional tests on the cognitive axes. Δ is the level-2-minus-level-0 change in the metric. All tests pass.

Axis	Metric	Δ (L2–L0)	$ \Delta $ bar	Expected sign	Pass
Contingent reasoning	truthful rate	+0.230	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	0.05	negative	✓
Forward planning	truthful rate	+0.137	0.05	positive	✓
Forward planning	underbid rate	−0.137	0.05	negative	✓
Belief reasoning	truthful rate	+0.052	0.03	positive	✓

Notes. Sign rate = 1.00 (5 of 5). Underlying level means (truthful rate): contingent $0.56 \rightarrow 0.86 \rightarrow 0.79$; forward $0.66 \rightarrow 0.75 \rightarrow 0.80$; beliefs $0.71 \rightarrow 0.74 \rightarrow 0.76$.

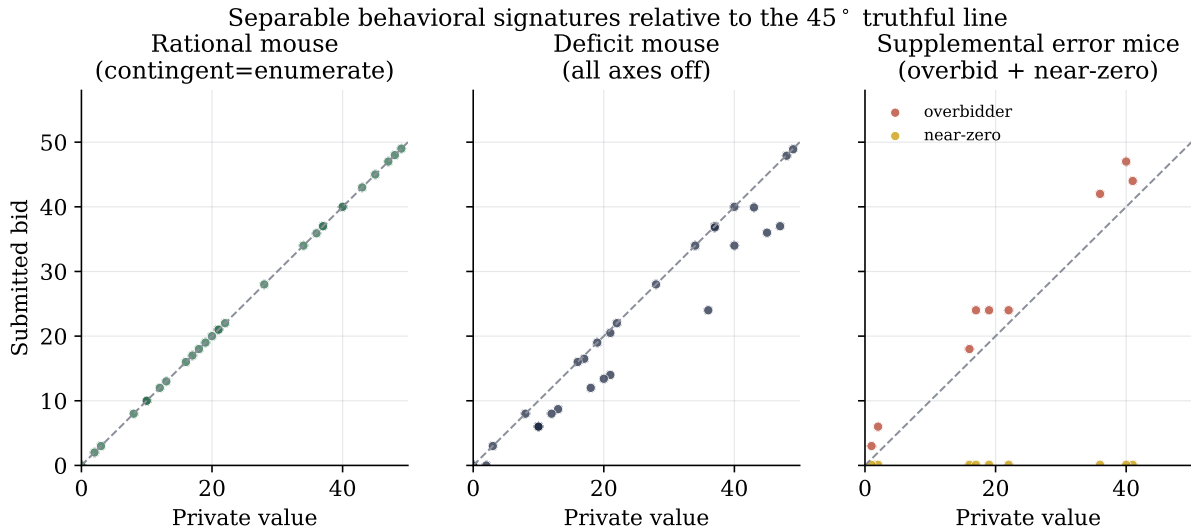


Figure 2: Separable behavioral signatures. Bid versus private value for a rational basis strain, the all-deficits-off strain, and the two supplemental error mice. The dashed line is the 45° truthful benchmark.

5 Result 2: Calibration Coverage

Construct validity establishes that the axes do what theory says. Calibration asks a different question: does the panel *span* the space of mistakes that real people make in this mechanism? We score five canonical SPSB human-mistake classes and ask, for each, whether some strain in the panel reproduces it at a rate above a detection threshold.

Figure 3 and Table 3 report the scorecard. All five classes are covered. Truthful and overbidding behavior are reproduced at very high rates (0.97 and 1.00); moderate underbidding, extreme overbidding, and near-zero bidding are reproduced at plausible intermediate rates (0.63, 0.44, 0.78). The best strain for each class is reported in the table: truthfulness and underbidding are covered by the clean procedural basis, whereas

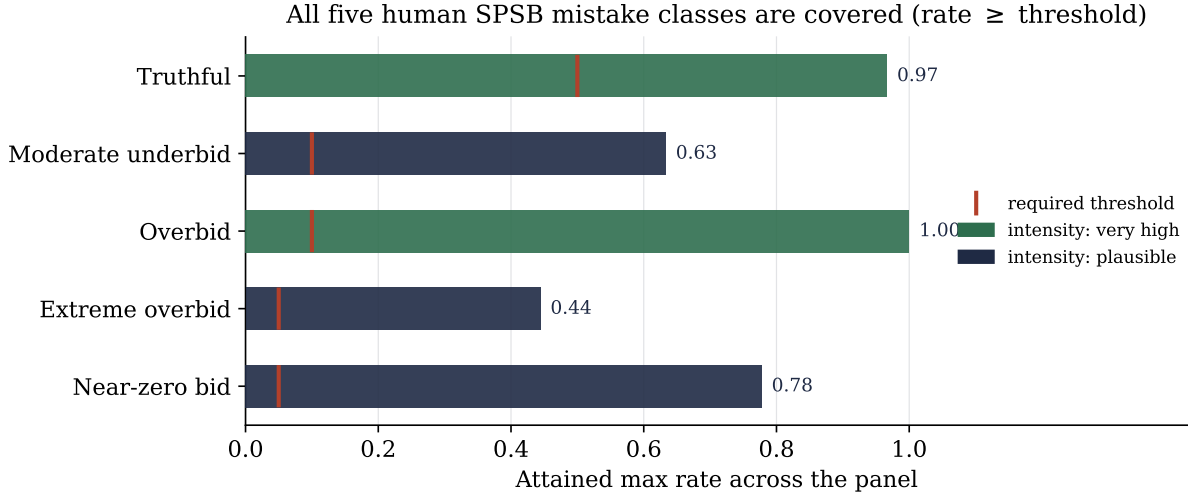


Figure 3: Calibration coverage. For each human-mistake class, the maximum rate attained by any strain in the panel (bar) against the required detection threshold (red marker). Bar color encodes the intensity band. All five classes clear their threshold.

Table 3: Calibration scorecard for the five canonical SPSB mistake classes. “Max rate” is the highest rate attained by any strain; “best strain” is the strain attaining it. All classes are covered.

Mistake class	Metric	Thresh.	Max rate	Best strain	Intensity	Covered
Truthful	truthful rate	0.50	0.967	c1_f0_b0	very high	✓
Moderate underbid	moderate underbid rate	0.10	0.633	c0_f0_b0	plausible	✓
Overbid	overbid rate	0.10	1.000	overgeneralizer [†]	very high	✓
Extreme overbid	extreme overbid rate	0.05	0.444	overgeneralizer [†]	plausible	✓
Near-zero bid	near-zero bid rate	0.05	0.778	payment panic [†]	plausible	✓

Notes. Coverage rate = 1.00 (5 of 5). [†] Supplemental error strain (spsb_error_second_price_overgeneralizer, spsb_error_payment_panic_near_zero); all other best strains are members of the clean 27-cell procedural basis.

the overbidding and near-zero classes are covered by the targeted error mice—making explicit which classes the procedural construction reaches on its own and which require hand-specified heuristics.

6 Result 3: Robustness Across Models and an All-Pay Stress Test

A panel is only useful for screening if its verdicts are not an artifact of the particular base model used to run it. We therefore put the panel to a harder test. We take a small set of

Table 4: Cross-model robustness of candidate mechanisms. Robust fitness and intended-action rate are computed against the same digital-mouse panel under two base models. Δ is held-out minus generating model.

Mechanism	Intended action	Robust fitness		Δ	Intended-action rate	
		mini	gpt-5.4		mini	gpt-5.4
all-pay: deposit, no reveal	bid aggressively	2.52	1.53	-1.00	0.63	0.30
all-pay: deposit, post-reveal	bid aggressively	2.85	1.27	-1.57	0.67	0.23
all-pay: reserve, pre-bid signal	mechanism-specific	-1.37	1.17	+2.54	—	—
second-price: deposit, binary	truthful	2.24	0.99	-1.25	0.60	0.60

Notes. “mini” is **gpt-5.4-mini** (the model used to generate the candidates); “gpt-5.4” is the held-out model. The reserve/pre-bid-signal mechanism declares a mechanism-specific target that the first-pass scorer does not evaluate, so its intended-action rate is omitted.

four candidate auction mechanisms—three all-pay variants (with a refundable deposit and varying information rules, and one with a reserve and a pre-bid signal) and one second-price design—and score each one against the digital-mouse panel under *two* base models: the model used to generate the candidates (**gpt-5.4-mini**) and a stronger held-out model (**gpt-5.4**). For each mechanism we declare the intended action of a correctly reasoning bidder—“bid aggressively” for the all-pay designs (a bidder who pays regardless of winning should bid up toward value), “truthful” for the second-price design—and measure the fraction of bidder actions that match it, together with a robust panel-fitness score.

Figure 4 and Table 4 report the result. Two facts stand out. First, robust fitness does not fully port: the two refundable-deposit all-pay mechanisms lose 1.0 and 1.6 fitness points when moved to the held-out model, and even the second-price design loses 1.3. Second, and more diagnostically, the all-pay intended action collapses on the stronger model: the “bid aggressively” rate falls from 0.63 to 0.30 and from 0.67 to 0.23, whereas the second-price “truthful” rate is unchanged ($0.60 \rightarrow 0.60$). In other words, a mechanism that looked robust under the generating model is substantially less so under a more capable one—a concrete instance of the overfitting risk that any model-based screening procedure must guard against. The stability of the truthful target relative to the aggressive-bidding target also mirrors the construct-validation finding that truthfulness is the most reliably induced behavior in this environment.

Strain-level portability. The mechanism-level gap raises a prior question: do the deficit *strains* themselves behave consistently across models? A pilot exposed a failure mode—the stronger model spontaneously “repaired” the most deficient basis strain into truthful Vickrey bidding—so we hardened the panel into a *strict* scaffold that states

Cross-model portability of candidate mechanisms on the digital-mouse panel

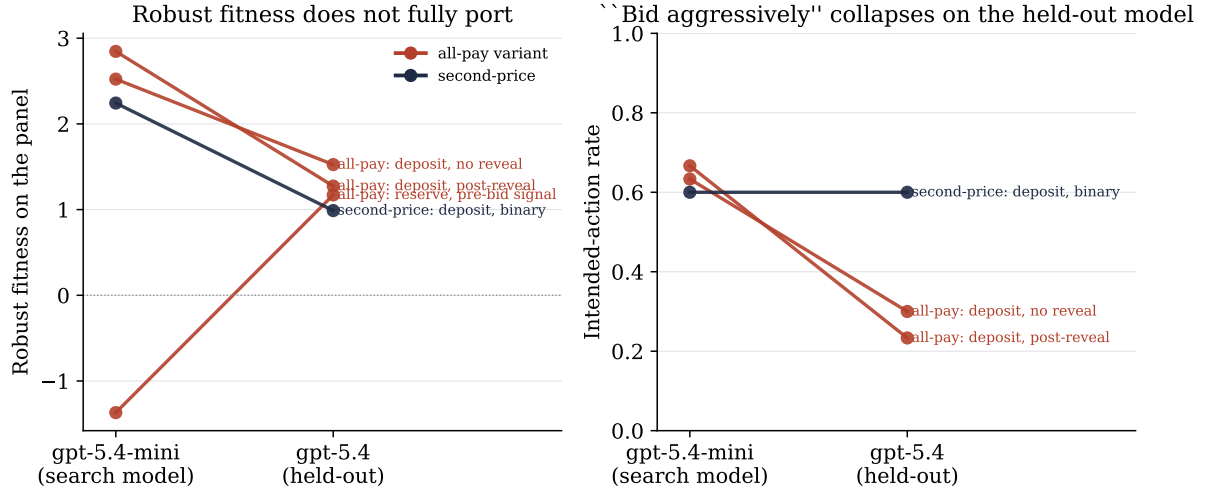


Figure 4: Cross-model portability of four candidate mechanisms on the digital-mouse panel. Left: robust panel-fitness under the generating model (**gpt-5.4-mini**) versus a held-out model (**gpt-5.4**). Right: fraction of bids matching the mechanism’s intended action. The all-pay “bid aggressively” target degrades sharply on the held-out model; the second-price truthful target is stable.

the reasoning-procedure constraint explicitly rather than as a secondary goal. We then re-ran five reserved strict strains under both base models and compared their dominant behavioral class (Table 5). Four of five strains preserve their dominant class across models—a partial pass. The exception is the strain isolating a contingent-reasoning deficit, whose moderate-underbidding signature on **gpt-5.4-mini** is repaired into truthful bidding on **gpt-5.4** (truthful rate $0.33 \rightarrow 1.00$). Consistent with the construct-validation finding that contingent reasoning is the highest-leverage axis (Table 2), it is also the hardest deficit to hold fixed against a more capable model: the strict scaffold mitigates the portability problem but does not fully close it.

7 Discussion

Procedural basis versus authored error mice. The construct-validity result is strongest for the procedural basis, where a single instruction (repair an axis) produces the predicted directional movement. The calibration result is more nuanced: the basis covers truthful and underbidding behavior on its own, but overbidding—a first-order regularity in human auction data (Kagel and Levin, 1993)—is recovered only with hand-specified heuristic strains. This distinction matters for interpretation. The model-agnostic claim

Table 5: Strain-level cross-model portability on the strict panel. Dominant behavioral class of five reserved strict strains under each base model. Four of five agree; the contingent-isolated strain’s deficit is repaired away by the stronger model.

Strain	Isolated role	mini → gpt-5.4 (dominant class)	Agree
c2_f2_b0	belief deficit	truthful → truthful	✓
c2_f0_b2	forward deficit	mod. underbid → mod. underbid	✓
c0_f0_b1	near-all-defect	mod. underbid → mod. underbid	✓
c1_f1_b1	all-intermediate	truthful → truthful	✓
c0_f2_b2	contingent deficit	mod. underbid → truthful	—

Notes. Agreement = 4/5. “Isolated role” names the axis held in its deficit state (others repaired), or a corner strain. The contingent-deficit strain’s truthful rate rises 0.33 → 1.00 across models; all other strains’ dominant class is stable.

applies cleanly to the procedural axes; the error mice are a complementary, more bespoke device for spanning the mistake space. The two should be reported as distinct epistemic objects.

Model dependence. Because deficits are defined as procedural controls rather than artifacts of one model, the protocol can be rerun on a different frontier model. Section 6 exploits this at two levels. At the mechanism level, the all-pay intended action degrades sharply on a held-out model. At the strain level, four of five reserved strict strains preserve their dominant behavioral class across models, the lone exception being the contingent-reasoning deficit, which a stronger model tends to repair into truthful bidding. Both findings are features of the design rather than bugs—the panel is precisely the instrument that makes such overfitting visible—but they caution against reading single-model panel scores as model-invariant, and they identify contingent reasoning as the deficit most in need of a hardened scaffold.

External validity. Digital mice are a phase-I screening population. A strain reproducing a mistake class establishes that the panel can express that behavior, not that the human share of that behavior is matched. We therefore keep the stress-testing claim (this paper) separate from any calibrated prediction of human behavior, which would require human pilot data.

8 Conclusion

We show that a frontier reasoning model can be instructed into stable, separable cognitive deficits whose behavior in a second-price sealed-bid auction (i) moves in the theory-

predicted direction as each cognitive axis is repaired and (ii) spans the canonical space of human SPSB mistakes once a small set of targeted error strains is added, and (iii) discriminates candidate mechanisms by robustness, including an all-pay stress test in which the intended action fails to port to a stronger held-out model. This supports a research program in which instructed language-model agents are used as a phase-I population to screen mechanisms for robustness to bounded rationality before human-subject experiments are run. The immediate next steps are to scale the cross-model robustness analysis beyond the present handful of mechanisms and to extend both the construct-validation and calibration protocols to a second domain (matching).

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